

Unit - 1

Year/Period	Major Development
1980s	Basic built-in applications (Calculator, Calendar)
1990s	PDA's, Symbian, Palm OS
2000–2006	Java ME, Feature phone applications
2007	Apple iPhone introduced
2008	App Store and Android launched
2010–2015	Rapid growth of smartphones and mobile apps
2015–2020	Cross-platform frameworks (Flutter, React Native)
2020–Present	AI, Cloud, IoT, AR/VR, 5G, wearable and foldable apps

1. Birth of Android (2003)

- **Android Inc.** was founded in **2003** by **Andy Rubin, Rich Miner, Nick Sears, and Chris White**.
- Initially, Android was intended to develop software for **digital cameras**, but the focus soon shifted to **mobile phones**.

2. Google Acquires Android (2005)

- In **2005**, **Google** acquired Android Inc.
- Google began developing Android as an **open-source mobile operating system** based on the **Linux kernel**.
- The goal was to create a flexible platform for smartphone manufacturers and developers.

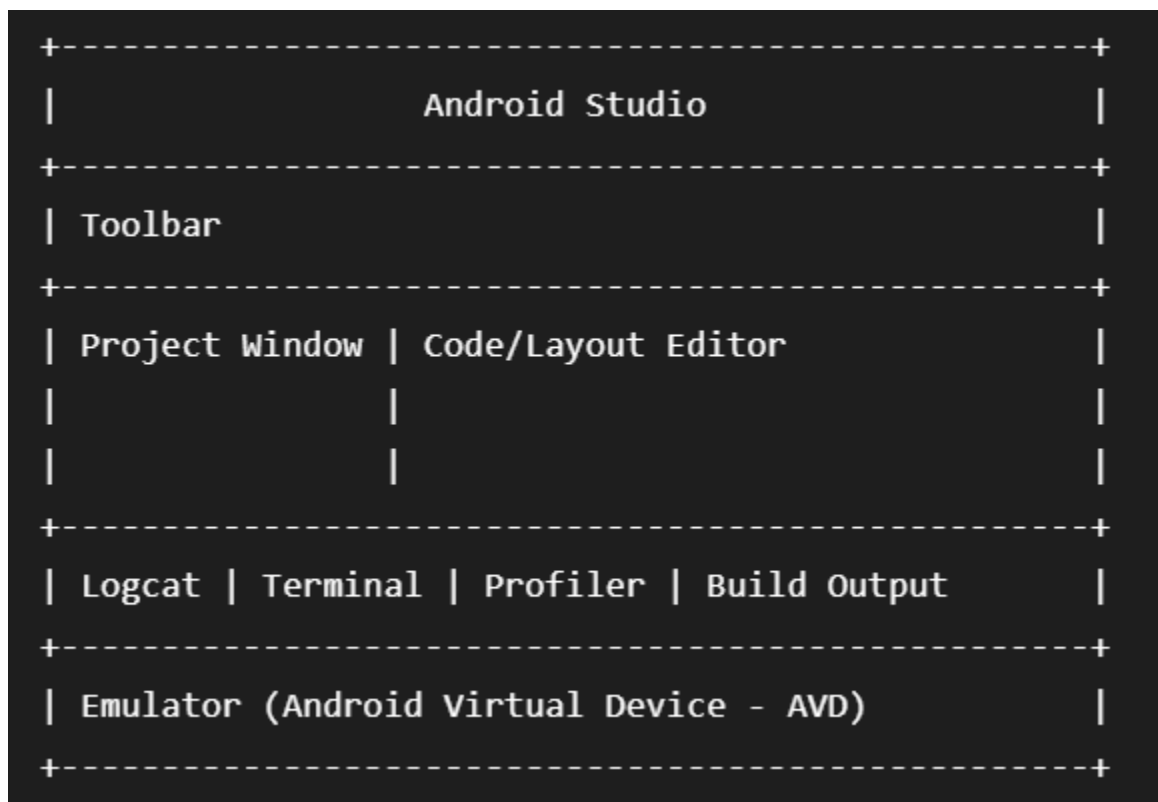
3. Formation of the Open Handset Alliance (2007)

- In **2007**, Google formed the **Open Handset Alliance (OHA)** with hardware manufacturers, software companies, and mobile carriers.
- The alliance aimed to develop open standards for mobile devices.
- Android was officially announced as an open-source platform.

4. First Android Phone (2008)

- The **HTC Dream (T-Mobile G1)** became the first Android smartphone in **2008**.
- Google also launched the **Android Market** (now **Google Play Store**) for downloading applications.
- Features included:
 - Touchscreen interface
 - Web browser
 - Gmail integration
 - Google Maps

Components of Android IDE



Activity: An Activity is a component of an Android application that represents a single screen and manages user interaction with that screen.

Lifecycle Methods

1. onCreate()

- Called when the activity is created.
- Initializes the UI and variables.
- Usually calls `setContentView()`.

2. **onStart()**

- Activity becomes visible to the user.

3. **onResume()**

- Activity comes to the foreground.
- User can interact with it.

4. **onPause()**

- Called when another activity partially covers the current one.
- Save temporary data and pause ongoing tasks.

5. **onStop()**

- Activity is no longer visible.
- Release unnecessary resources.

6. **onRestart()**

- Called when the activity is restarted after being stopped.

7. **onDestroy()**

- Called before the activity is destroyed.
- Clean up resources and perform final tasks.

Ex: public class MainActivity extends AppCompatActivity {

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

An **Intent** is a **messaging object** used to request an action, such as starting another activity or passing data between activities.

Set up android environment

Check system requirements.

Download and install Android Studio.

Run the Setup Wizard.
Install the Android SDK.
Configure the SDK Manager.
Create an Android Virtual Device (AVD).
Create a new Android project.
Build, run, test, and debug the application.

Android Lifecycle Code

```
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        System.out.println("onCreate");
    }

    @Override
    protected void onStart() {
        super.onStart();
        System.out.println("onStart");
    }

    @Override
    protected void onResume() {
        super.onResume();
        System.out.println("onResume");
    }

    @Override
    protected void onPause() {
        super.onPause();
        System.out.println("onPause");
    }

    @Override
    protected void onStop() {
        super.onStop();
        System.out.println("onStop");
    }
}
```

```

@Override
protected void onRestart() {
    super.onRestart();
    System.out.println("onRestart");
}

@Override
protected void onDestroy() {
    super.onDestroy();
    System.out.println("onDestroy");
}
}

```

A **View** is the basic building block of the Android User Interface (UI). It is a **single UI component** that displays content or allows the user to interact with the application.

A **ViewGroup** is a special type of **View** that acts as a **container** for other Views and ViewGroups. It is used to organize and arrange UI components on the screen.

Types Of Layout :

LinearLayout
 RelativeLayout
 ConstraintLayout
 FrameLayout
 TableLayout
 GridLayout

View	ViewGroup
Basic UI element.	Container that holds Views and other ViewGroups.
Cannot contain child views.	Can contain multiple child views.
Used to display data or receive user input.	Used to arrange and organize UI elements.
Examples: Button, TextView, EditText, ImageView.	Examples: LinearLayout, ConstraintLayout, FrameLayout, RelativeLayout.
Inherits from the View class.	Also inherits from View , but is designed to manage child views.

Thread

A class (`java.lang.Thread`).

Represents an actual thread of execution.

Created by extending the `Thread` class.

Starts execution using `start()`.

Cannot extend another class if it already extends `Thread`.

Less reusable.

Suitable for simple tasks.

Runnable

An interface (`java.lang.Runnable`).

Represents a task to be executed.

Implemented by implementing the `Runnable` interface.

Requires a `Thread` object to execute.

Allows the class to extend another class because it only implements an interface.

More reusable and flexible.

Preferred for most Android applications.